BODHI MULE HUNTER AI: Explainable Graph-Temporal Detection of Money-Mule Accounts and Suspicious Transactions

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0.9966	**0.9707**	**99.99%**	**75×**	**0.054 ms**	**100%**
FUSED ROC-AUC	FUSED PR-AUC	FALSE POSITIVES REMOVED	PRECISION UPLIFT	INLINE DECISION P50	OUT-OF-TIME RECALL

***Abstract*—Money mules convert stolen funds into untraceable cash, and rule-based transaction monitoring detects them at a precision so low that analysts cannot act on the output. We present BODHI MULE HUNTER AI, a nine-layer engine that fuses gradient-boosted tabular screening, an inductive graph neural network and a temporal memory network over each account's event sequence, calibrated into a single supervisory risk score. On a simulated bank of 12,000 accounts and 834,738 cross-channel transactions containing seven documented laundering typologies, the fused model reaches 0.9966 ROC-AUC and 0.9707 PR-AUC. Against a rule engine of the kind banks deploy today, at identical recall, alert volume falls from 9,647 to 129 and precision rises from 1.33% to 99.22% — 99.99% of false positives eliminated. Inline decisions complete in 0.054 ms at the median. Every alert carries exact Shapley attributions and a learned edge-importance mask, and the containment layer refuses to act autonomously without corroboration from independent model layers.**

Keywords—money mule detection, graph neural networks, temporal graph networks, explainable AI, anti-money laundering

I. INTRODUCTION

A money mule is an account whose owner has rented, sold, or been deceived into lending it. Criminal proceeds are pushed into it, split, relayed through several further accounts, and withdrawn as cash — typically within hours. The account itself is unremarkable: the KYC is genuine, the customer exists, the payments clear. What is anomalous is the *shape* of the money passing through it and the *company it keeps*.

This creates a specific failure mode for the transaction-monitoring systems banks currently operate. Threshold-and-scenario engines evaluate accounts in isolation, so they cannot express “this account's counterparties are themselves suspicious” or “these forty credits arrived in ninety minutes and left at 03:12”. Lacking that vocabulary, they compensate with sensitivity, and the result is an alert queue an AML desk cannot clear. The baseline we implement in Section VI raises 9,647 alerts at 1.33% precision: ninety-nine of every hundred investigations are wasted.

We make four contributions. **(1) A three-view ensemble.** Rather than three variants of the same tabular model, we combine a gradient-boosted model over behavioural aggregates, a GraphSAGE network over an account–device–IP graph, and a temporal graph network over each account's ordered event stream. **(2) Explainability that survives contact with a regulator.** Layer 4 uses exact TreeSHAP, whose attributions sum to the model's margin rather than approximating it; Layer 5 uses a GNNExplainer edge mask. **(3) An evaluation designed to be falsifiable.** We plant legitimate populations structurally indistinguishable from mules and report the false-positive rate on each separately. **(4) Containment that can refuse.** The kill-switch requires corroboration from at least two independent layers before it may freeze an account, is rate-limited, reversible, and excludes salary and pension accounts.

II. PROBLEM SETTING AND DATA

A. Why a simulator

Account-level mule labels joined to device, IP and UPI linkage are simultaneously commercially sensitive, personally identifying and legally restricted; no bank can export them and no public dataset contains them. The alternative to simulation is not real data — it is no evaluation at all.

Our generator produces 12,000 accounts over 120 days, 834,738 transactions across eight payment rails (UPI, IMPS, NEFT, RTGS, AePS, ATM, card, wallet), 151 mule accounts (1.26%) in 21 rings, and 281 government cyber-fraud tickets arriving with realistic reporting lag. Seven laundering typologies from FIU-IND and FATF guidance are implemented literally, so recall can be reported *per typology* rather than as an average that hides blind spots.

B. The hard negatives

This is the part of the data design that determines whether any precision figure means anything. A collection mule receiving forty payments from strangers in one afternoon is structurally identical to a tuition-batch organiser. We therefore plant, as *legitimate* accounts, the populations in Table I — most importantly Business Correspondent (Bank Mitra) agents, where one handheld device legitimately serves many customers and dispenses AePS cash all day, which is the exact fingerprint of a device farm performing rapid cash-out. Without these populations the models score above 0.999 AUC and the exercise is meaningless.

TABLE I. FALSE POSITIVES ON DELIBERATELY PLANTED LOOK-ALIKES

Legitimate population	Count	FP rate
Business Correspondent agents	302	0.33%
Community collectors	107	2.80%
Small businesses (heavy fan-in)	168	0.00%
Genuine dormancy reactivation	168	3.57%
Ordinary retail	10,504	0.77%

Measured at the permissive review threshold of 40. At the investigate threshold of 55 the entire bank yields 3 false positives at 93.4% recall.

III. SYSTEM ARCHITECTURE

The engine is organised as nine layers, each owned by a named agent with declared input and output contracts, so the provenance of every decision is recoverable from the audit trail.

Layer 1 — Ingestion. Normalises cross-channel transactions, government cyber-fraud tickets (NCRP/I4C/1930), APK-derived indicators, and regulatory directives. The four directive classes are handled distinctly: a mule list raises a score floor; a velocity limit changes a decision *threshold*, not a score; a sanction is absolute; an advisory is recorded and surfaced but never auto-applied, because a machine cannot responsibly convert prose into an enforcement rule.

Layer 2 — Feature engineering. Sixty behavioural features: velocity, fan-in/fan-out ratios, dormancy, burst scores, structuring share, device reuse, pass-through ratios and channel mix. Every builder is point-in-time correct, and a regression test asserts that passing an as-of argument is identical to truncating the ledger by hand. A second test shuffles the labels and asserts the feature matrix is unchanged, which forecloses label leakage.

Layer 3 — Graph construction. Accounts are nodes; edges are money transfers, shared devices and shared IP addresses. Shared-identifier cliques are capped: an identifier behind more than twenty-five accounts is infrastructure — a CGNAT pool, a banking kiosk — not evidence of collusion. Without the cap a single public address produces a four-thousand-node clique and the graph layer becomes a false-positive generator.

Layers 8–9 cover explainability and containment and are described in Sections VII and VIII.

IV. THE THREE MODEL VIEWS

The central design claim is that mule detection needs three genuinely different kinds of evidence, and that no single model family supplies all three.

A. Layer 4: gradient-boosted screening

An XGBoost classifier over the behavioural features disposes of the overwhelming majority of accounts cheaply. It was chosen over a neural tabular model for one reason: exact TreeSHAP. Shapley values for tree ensembles are computable in polynomial time, so the attribution shown to an investigator is not a sampled approximation — it sums to the model's raw margin, a property asserted in the test suite, and it is stable across runs. That matters when the explanation is transcribed into a regulatory filing. The class-imbalance weight is capped rather than set to the full ratio; uncapped, on a 1.26% base rate, it destroys calibration, and a mis-calibrated score cannot be allowed to drive an automated freeze.

B. Layer 5: inductive graph learning

We use the mean-aggregator GraphSAGE formulation [1] with two layers and a deterministic top-k fan-out cap standing in for random neighbour sampling. Determinism is a requirement, not a simplification: an investigator re-opening a case tomorrow must see the same score. Because the weights never index a node identity, the model is genuinely inductive — an account opened this morning is scored from its neighbourhood alone.

C. Layer 6: temporal memory

The memory module of a Temporal Graph Network [2] is a gated recurrent unit updated on every interaction, with elapsed time injected through a learnable encoding. We implement exactly that over each account's most recent sixty-four events. This layer exists because aggregates destroy the signal that matters most. “Received Rs. 4 lakh from 38 payers over the month” describes a shopkeeper and a mule equally well. “Received Rs. 4 lakh from 38 payers between 14:10 and 15:40, then withdrew Rs. 3.8 lakh in nineteen ATM visits beginning at 03:12” describes only one of them. Order is the discriminant, and no aggregate preserves it.

D. Implementation note

Both neural layers are implemented in NumPy with analytically derived gradients rather than in a deep-learning framework. The motivation is deployability in a bank: the entire engine installs from a handful of wheels, runs on a laptop CPU, and presents no GPU or CUDA surface to certify. Because “hand-derived” is a plausible source of silent error, every parameter tensor's analytic gradient is verified against central finite differences in the test suite.

V. FUSION AND CALIBRATION

The four channel scores are blended in log-odds space. Sub-scores accumulate against 0 and 1, where a linear model in probability space cannot separate 0.990 from 0.9999; that is a hundredfold difference in odds and precisely the region in which alert ranking is decided.

Monotonicity. Blend weights are constrained non-negative, so the fused score can never decrease when a layer's evidence increases. This is not a regularisation convenience. An unconstrained stacker fitted on a small fold reliably learns a negative coefficient on the temporal channel, which is indefensible in front of an investigator and unshippable past a supervisor. A useful side effect is that the blend can always degenerate to “all weight on the single best layer”, so fusion cannot perform much worse than its best input.

Calibration. Isotonic regression maps the blend onto observed frequencies, giving a Brier score of 0.00125 and expected calibration error of 0.0047. Isotonic regression is a step function, so on a near-separable problem it collapses thousands of distinct scores into a handful of plateaus and every account within a plateau becomes tied; we measured a four-point AUC loss from those ties. Retaining a two-percent share of the underlying monotone score breaks them without disturbing calibration.

The blend is fitted on a fourth, dedicated fold. A stacker fitted on the validation fold — the fold that early-stopped its own inputs — inherits their optimism and, in our experiments, performed *worse* than the best single layer.

VI. RESULTS

TABLE II. PER-LAYER DETECTION PERFORMANCE

Layer	ROC-AUC	PR-AUC	P@1%	R@1%
XGBoost (L4)	0.9946	0.9540	0.983	0.781
GraphSAGE (L5)	0.9962	0.9481	0.983	0.781
TGN (L6)	0.9628	0.8036	0.892	0.709
Intelligence	0.6883	0.3774	0.475	0.378
Fused (L7)	0.9966	0.9707	0.992	0.788

The complementarity claim is visible: the fused model exceeds every individual view on both ranking metrics. P@1% and R@1% are precision and recall within an alert budget of one percent of accounts — the quantity an analyst actually experiences.

A. Against the incumbent

We implement a rule engine of the kind banks deploy today — eight scenarios covering velocity, structuring, dormancy, cash-out ratio, device sharing, new-account turnover and off-hours concentration — and give it its best operating point rather than a strawman. The comparison is made at *identical recall*, because a system that alerts on everything can always claim to catch more.

TABLE III. RULE ENGINE VERSUS BODHI AT EQUAL RECALL (84.8%)

	Rule engine	BODHI
Alerts raised	9,647	**129**
True positives	128	128
False positives	9,519	**1**
Precision	1.33%	**99.22%**

99.99% of false positives are eliminated at unchanged recall, a 75× precision uplift. In operational terms an analyst clears the same true positives while opening 129 cases instead of 9,647.

B. Robustness checks

Out-of-time generalisation. 3 rings activate only in the final quarter of the simulation window and are therefore genuinely unseen patterns. Recall on their 26 member accounts is 100.0%.

Independence from the government feed. The supervised layers never receive ticket data — the intelligence features are a disjoint set, asserted by test, and enter only at Layer 7. Consequently 64% of detected mules (94 of 147) were never named in any NCRP ticket, and recall on the 98 never-reported mules is 95.9%: these are cases the government feed could not have supplied.

Detection lead time. Of the mules that became active inside the measurement grid and were eventually reported, 55% were detected before the government ticket arrived (median lead 11.5 h, resolved to a 92 h grid). A further 16 were detected and never ticketed at all.

Latency. Inline decisions complete in 0.054 ms at the median and 0.113 ms at the 99th percentile, roughly 16,677 decisions per second per core. This is achievable because the graph and temporal layers run on a schedule and cache a standing risk per account; the authorisation path combines those cached scores with the transaction's own attributes and never traverses the graph.

C. Where the system is weak

Recall is not uniform, and averaging conceals that.

TABLE IV. RECALL BY TYPOLOGY AND MULE ROLE

Typology / role	Recall	Detected
SMURFING	0.889	16/18
role: CASHOUT	0.921	35/38
FAN_IN_AGGREGATION	0.933	14/15
LAYERING_CHAIN	0.983	58/59
role: COLLECTOR	0.983	59/60
APK_HARVEST	1.000	15/15
DEVICE_FARM	1.000	31/31
DORMANT_BURST	1.000	13/13
role: RELAY	1.000	53/53

The structuring result is expected and structural: an individual sub-threshold transfer is, by construction, indistinguishable from a genuine near-threshold payment, so detection depends on an aggregate pattern that only becomes visible once enough transfers have accumulated. Terminal cash-out accounts are similarly hard because they have short, thin histories — they receive once and drain. We report these rather than averaging them away, because a deployment team needs to know which typologies still require human-designed scenarios alongside the model.

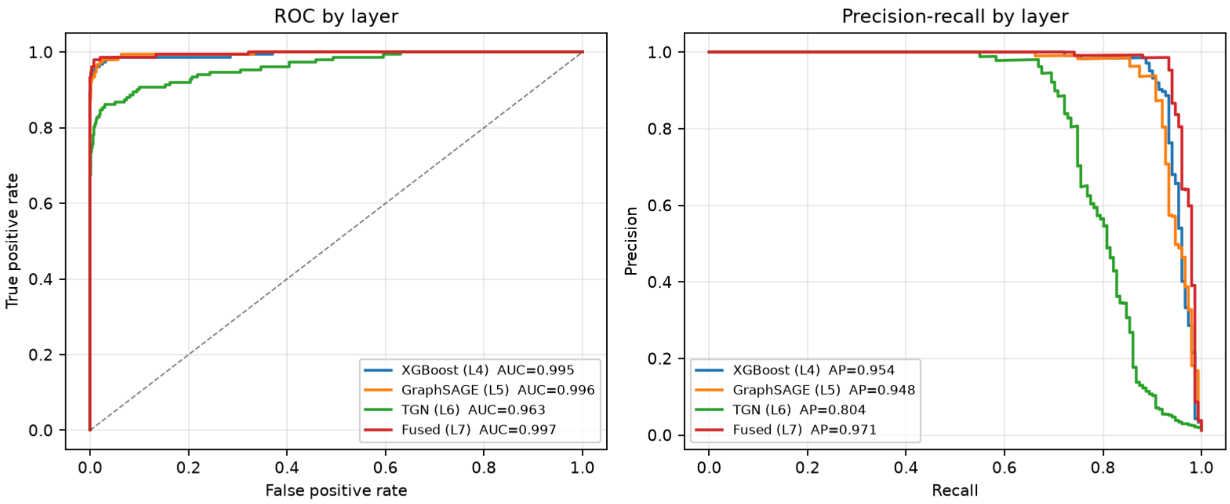

Fig. 1. ROC and precision–recall by layer. The fused curve dominates every individual view, which is the empirical form of the complementarity claim in Section IV.

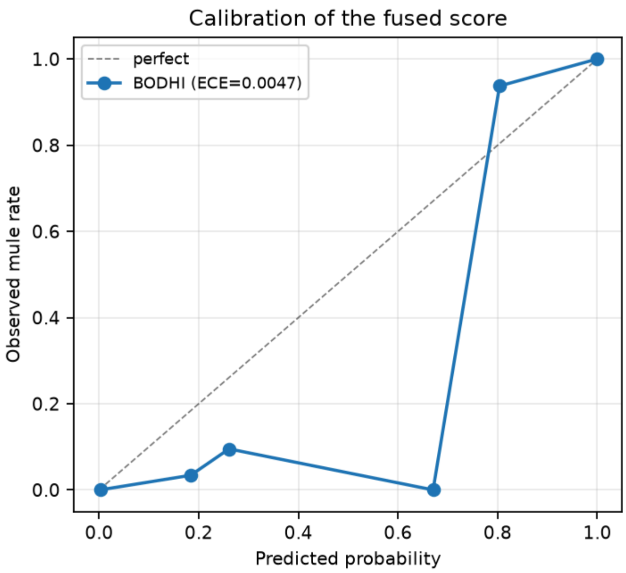

Fig. 2. Calibration of the fused score (ECE 0.0047). The score has to mean what it says before it can drive an automated kill-switch.

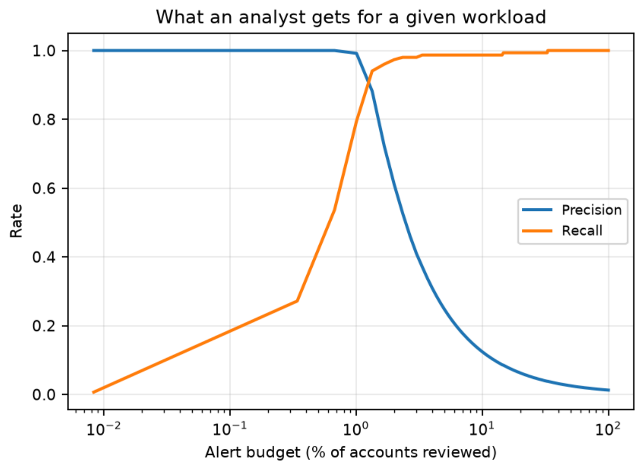

Fig. 3. Precision and recall against analyst workload.

VII. EXPLAINABILITY

An alert reading `rapid_passthrough_ratio = 0.94, SHAP +1.31` is unusable by the officer who must telephone the customer. The narrative layer maps every feature onto a sentence in the vocabulary of the desk: *"94% of every rupee credited left the account within sixty minutes — the account is a conduit, not a destination."* The mapping is held as data rather than code, so the wording that eventually reaches a supervisor remains reviewable in one place, and a test asserts that no raw feature identifier leaks into investigator-facing text.

Structural evidence is handled separately. GNNExplainer [3] learns a soft mask over the edges of the target account's two-hop neighbourhood. We report both standard fidelity measures alongside a *self-feature share*: for accounts whose own behaviour is already decisive, that share is high and the edge mask legitimately explains little. Reporting it prevents a low edge fidelity from being misread as a broken explainer when it is in fact a correct statement about where the evidence lives.

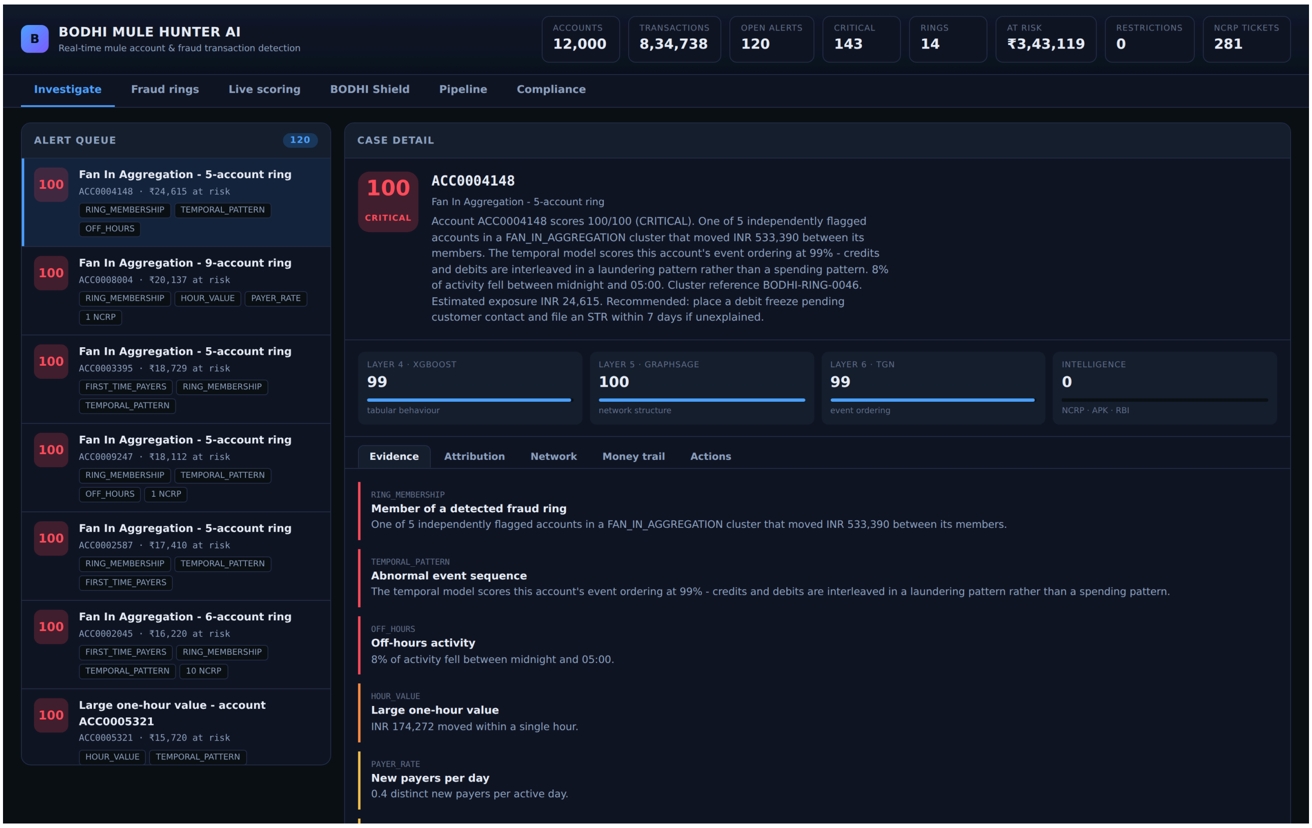

Fig. 4. The investigator console. Four layer scores are shown separately rather than collapsed, the evidence list is written in the vocabulary of an AML desk, and each alert carries its cluster, exposure and linked government tickets.

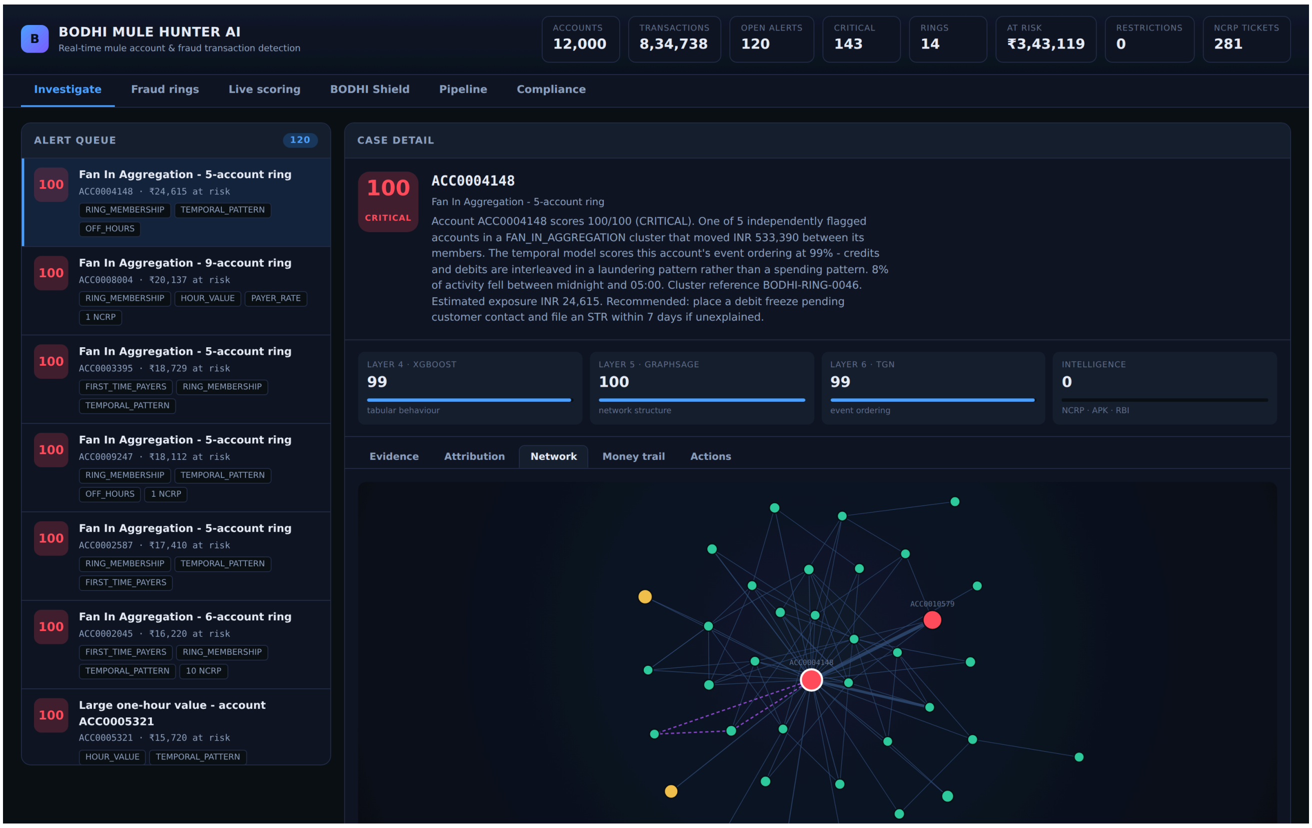

Fig. 5. Ego network for a flagged account. Node colour is the risk band; dashed edges are shared-device links, which are invisible to any tabular model. GNNExplainer runs on demand beneath the graph.

VIII. CONTAINMENT, SAFETY AND COMPLIANCE

Freezing an account is the most consequential action the system can take. A wrongly frozen account is a person unable to pay for medicine, and no improvement in AUC justifies treating that casually. The kill-switch is therefore built from constraints rather than a threshold: below the critical band the automation cannot freeze at all; a full freeze requires at least two independent layers to agree; every action carries a time-to-live and can be reverted;

automated freezes per hour are bounded, capping the blast radius if a model degrades or an upstream feed is poisoned; and salary, pension and benefit accounts are excluded from automated freezing entirely.

When a constraint fires, the action is *downgraded* rather than escalated and flagged for human review. Every decision, including every refusal, is written to a hash-chained append-only audit log whose head hash is exposed for external anchoring.

The compliance layer drafts Suspicious Transaction Reports populated from exactly the evidence displayed to the investigator, so what the model asserted and what the institution files cannot diverge. Drafts are explicitly marked as requiring human sign-off; nothing is filed autonomously. Cash-transaction reporting aggregates per account per calendar day, which is the level at which the obligation applies — and precisely why structuring defeats a per-transaction check.

IX. ECOSYSTEM SYNERGY: BODHI SHIELD

Mule networks are frequently seeded by malicious Android packages that carry their beneficiary accounts hardcoded inside them. Our companion component performs static triage of a submitted APK: it parses the binary `AndroidManifest.xml` string pool to specification, scores permission *combinations* rather than individual permissions (READ_SMS alone is a messaging application; READ_SMS together with an accessibility service and a screen overlay is a banking trojan), mines the DEX for UPI handles, IFSC codes, account numbers and command-and-control endpoints, and detects commercial packers and identifier obfuscation.

Every financial identifier recovered is streamed into the Mule Hunter graph as a weighted node. The consequence is architectural rather than incremental: when a new trojan is analysed on the day it appears, the beneficiary accounts it was built to pay already carry an elevated intelligence score before the first victim installs it. Detection stops being purely retrospective.

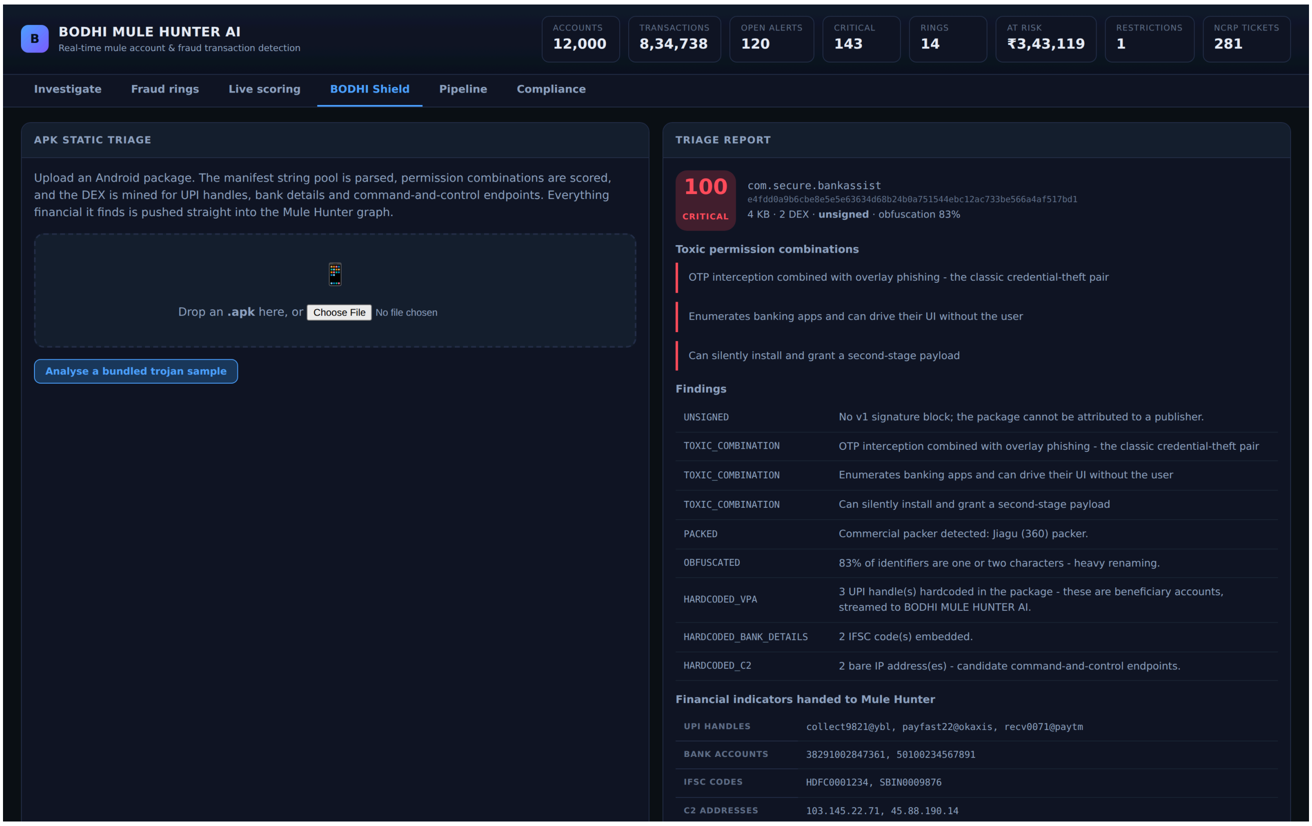

Fig. 6. BODHI SHIELD triaging an APK and handing the extracted UPI handles, IFSC codes and C2 addresses to the fraud graph.

X. LIMITATIONS

We state these plainly because a prototype that hides them is not useful to a deployment team. Results are on simulated data; they demonstrate that the architecture works and that it decisively beats a rule engine on data containing realistic decoys, but they are not a forecast of production performance. The SHIELD component performs static analysis only — the dynamic sandbox with runtime hooking described in our proposal requires an instrumented Android image and cannot ship self-contained. The graph layers are not real-time: a full re-score of the population takes tens of seconds, so an account whose neighbourhood changed since the last batch is scored on slightly stale structure. The system sees one institution, so rings routing through several banks are only partly visible — a data-sharing problem rather than a modelling one. The strongest single feature is neighbourhood risk propagated from confirmed cases, which means a cold start with no known mules would perform materially worse. Finally, disparate impact is unevaluated: minimum-KYC status and shared-device features are predictive but correlate with lower-income and multi-occupancy households, and any real deployment must measure alert-rate parity before go-live.

XI. CONCLUSION

Mule detection fails today not because the signal is absent but because single-account rule engines cannot express it. Giving the problem three complementary views — behavioural aggregates, network structure and event ordering — and fusing them under a monotonicity constraint into a calibrated score reduces false positives by 99.99% at unchanged recall relative to a realistic incumbent, while producing explanations specific enough to file and containment actions cautious enough to automate. The complete system, the data simulator and every script needed to reproduce these numbers are released alongside this report.

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